Parabola: Questions and Full Solutions

Question 01

English

Let E be the ellipse $\dfrac{x^2}{4} + y^2 = 1$ and let A be the point $(-2, 0)$. For a point P on the parabola $y^2 = 4x$, the line AP meets E again at the point Q. As P varies over the parabola, the least value of the abscissa of Q is

(A) $-\dfrac{1}{2}$

(B) $-\dfrac{1}{3}$

(C) $-\dfrac{2}{3}$

(D) $-\dfrac{7}{9}$

Only one option is correct.

हिन्दी

माना कि E दीर्घवृत्त $\dfrac{x^2}{4} + y^2 = 1$ है तथा A बिन्दु $(-2, 0)$ है। परवलय $y^2 = 4x$ पर स्थित किसी बिन्दु P के लिए, रेखा AP दीर्घवृत्त E को पुनः बिन्दु Q पर मिलती है। जब P परवलय पर गति करता है, तब Q के भुज का न्यूनतम मान है

(A) $-\dfrac{1}{2}$

(B) $-\dfrac{1}{3}$

(C) $-\dfrac{2}{3}$

(D) $-\dfrac{7}{9}$

केवल एक विकल्प सही है।

Solution

The ellipse is $\dfrac{x^2}{4} + \dfrac{y^2}{1} = 1$, so its points are $Q(2\cos\theta, \sin\theta)$, and $A(-2, 0)$ corresponds to $\theta = \pi$. The point Q is the second intersection, so $\theta \neq \pi$.

A point of the parabola $y^2 = 4x$ is $P(t^2, 2t)$, $t \in \mathbb{R}$. Since $t^2 \geq 0 > -2$, the line AP is never vertical, so it is never the tangent $x = -2$ at A and always meets E at a second point.

Slope of AQ: $\dfrac{\sin\theta - 0}{2\cos\theta + 2} = \dfrac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{4\cos^2\frac{\theta}{2}} = \dfrac{1}{2}\tan\dfrac{\theta}{2}$.

Slope of AP: $\dfrac{2t - 0}{t^2 + 2}$.

Since A, P, Q are collinear, $\dfrac{1}{2}\tan\dfrac{\theta}{2} = \dfrac{2t}{t^2 + 2}$, i.e. $T = \tan\dfrac{\theta}{2} = \dfrac{4t}{t^2 + 2}$.

By AM-GM, $t^2 + 2 \geq 2\sqrt{2}\,|t|$, so $|T| \leq \dfrac{4|t|}{2\sqrt{2}|t|} = \sqrt{2}$ for $t \neq 0$ (and $T = 0$ for $t = 0$). Hence $0 \leq T^2 \leq 2$, and $T^2 = 2$ exactly when $t^2 = 2$.

The abscissa of Q is $x_Q = 2\cos\theta = 2 \cdot \dfrac{1 - T^2}{1 + T^2} = 2\left(\dfrac{2}{1 + T^2} - 1\right)$, which is a decreasing function of T^2.

So x_Q is least when $T^2 = 2$: $x_Q = 2 \cdot \dfrac{1-2}{1+2} = -\dfrac{2}{3}$. This is attained for $P(2, \pm 2\sqrt{2})$, when $Q = \left(-\dfrac{2}{3}, \pm\dfrac{2\sqrt{2}}{3}\right)$; check: $\dfrac{4/9}{4} + \dfrac{8}{9} = 1$.

Correct Option: (C)

Question 02

English

A circle with centre $C(-12, 0)$ cuts the parabola $y^2 = 12x$ orthogonally. Then the radius of the circle is

(A) $12\sqrt{5}$
(B) $6\sqrt{5}$
(C) 24
(D) $12\sqrt{2}$

Only one option is correct.

हिन्दी

केन्द्र $C(-12, 0)$ वाला एक वृत्त परवलय $y^2 = 12x$ को लम्बकोणीय रूप से (orthogonally) काटता है। तब वृत्त की त्रिज्या है

(A) $12\sqrt{5}$
(B) $6\sqrt{5}$
(C) 24
(D) $12\sqrt{2}$

केवल एक विकल्प सही है।

Solution

Let the circle meet the parabola at P. The tangent to the circle at P is perpendicular to the radius CP.

The curves cut orthogonally at P, so the tangent to the circle at P is perpendicular to the tangent to the parabola at P. Two lines through P perpendicular to the same line coincide, hence the line CP is the tangent to the parabola at P.

For $y^2 = 12x$ we have $a = 3$, so $P = (3t^2, 6t)$ and the tangent at P is $ty = x + 3t^2$.

This tangent passes through $C(-12, 0)$: $0 = -12 + 3t^2$, so $t^2 = 4$, $t = \pm 2$, and $P = (12, \pm 12)$.

Radius $r = CP = \sqrt{(12+12)^2 + (12-0)^2} = \sqrt{576 + 144} = \sqrt{720} = 12\sqrt{5}$.

Check that the configuration is consistent: the circle $(x+12)^2 + y^2 = 720$ meets $y^2 = 12x$ where $x^2 + 24x + 144 + 12x = 720$, i.e. $x^2 + 36x - 576 = 0$, i.e. $(x-12)(x+48) = 0$. Since $x \geq 0$ on the parabola, $x = 12$, giving only $(12, 12)$ and $(12, -12)$, and at both the slope of CP $(\pm\frac{1}{2})$ equals the slope of the parabola's tangent $(\frac{6}{y} = \pm\frac{1}{2})$.

Correct Option: (A)

Question 03

English

Let P be the point on the parabola $y^2 = 4x$ that is nearest to the point $A(-1, 14)$, and let S be the focus of the parabola. Then the area of the triangle SAP is

(A) 20
(B) 25
(C) 50
(D) 30

Only one option is correct.

हिन्दी

माना कि परवलय $y^2 = 4x$ पर स्थित P वह बिन्दु है जो बिन्दु $A(-1, 14)$ के निकटतम है, तथा S परवलय की नाभि है। तब त्रिभुज SAP का क्षेत्रफल है

(A) 20
(B) 25
(C) 50
(D) 30

केवल एक विकल्प सही है।

Solution

For a point $P(t^2, 2t)$ of $y^2 = 4x$, the square of the distance from $A(-1, 14)$ is $f(t) = (t^2 + 1)^2 + (2t - 14)^2$.

$f'(t) = 4t(t^2 + 1) + 4(2t - 14) = 4(t^3 + 3t - 14)$. Equivalently, the nearest point is the foot of a normal through A: the normal at $(t^2, 2t)$ is $y = -tx + 2t + t^3$, and it passes through $(-1, 14)$ when $14 = t + 2t + t^3$, the same cubic.

$t^3 + 3t - 14 = (t - 2)(t^2 + 2t + 7)$, and $t^2 + 2t + 7 = (t + 1)^2 + 6 > 0$. So $t = 2$ is the only real root. Since $t^3 + 3t - 14$ is increasing, $f'(t) < 0$ for $t < 2$ and $f'(t) > 0$ for $t > 2$; hence f has its global minimum at $t = 2$ and the nearest point is $P(4, 4)$ (distance $\sqrt{25 + 100} = 5\sqrt{5}$).

The focus of $y^2 = 4x$ is $S(1, 0)$. With $\overrightarrow{SA} = (-2, 14)$ and $\overrightarrow{SP} = (3, 4)$, area of triangle $SAP = \frac{1}{2}|(-2)(4) - (14)(3)| = \frac{1}{2}|-8 - 42| = 25$.

Correct Option: (B)

Question 04

English

Let C be the parabola whose focus is the point $S(1,1)$ and whose directrix is the line $3x+4y+3=0$. Then the area of the region bounded by C and the line $3x+4y=47$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वह परवलय है जिसकी नाभि बिन्दु $S(1,1)$ है तथा जिसकी नियता रेखा $3x+4y+3=0$ है। तब C तथा रेखा $3x+4y=47$ से परिबद्ध क्षेत्र का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The axis of C is the line through S perpendicular to the directrix $3x+4y+3=0$, i.e. along the direction $(3,4)$.

Distance of $S(1,1)$ from the directrix: $\dfrac{|3(1)+4(1)+3|}{\sqrt{9+16}} = \dfrac{10}{5} = 2$. This distance is $2a$, so $a=1$; the latus rectum is $4a=4$ and the vertex lies on the axis at distance $a=1$ from the directrix.

The line $3x+4y=47$ is parallel to the directrix, so it is perpendicular to the axis. Its distance from the directrix is $\dfrac{|47-(-3)|}{5} = 10$. The focus gives $3x+4y=7$, which lies between -3 and 47, so the line is on the same side of the directrix as the focus, at distance $10-1=9$ from the vertex.

Distances are unchanged by a rotation and a translation, so in a frame with the vertex at the origin and the axis as the X-axis the parabola is $Y^2=4aX=4X$ and the line is $X=9$.

The line meets the parabola where $Y^2=36$, $Y=\pm 6$: the chord has length 12.

By Archimedes' result, the parabolic segment has area $\dfrac{2}{3}$ of the rectangle on the chord with height 9 (the tangent at the vertex is parallel to the chord): area $= \dfrac{2}{3}\cdot 12\cdot 9 = 72$.

(Check by integration: $2\displaystyle\int_0^9 2\sqrt{X}\,dX = 4\cdot\frac{2}{3}\cdot 27 = 72$.)

Final Answer: 72

Question 05

English

Let E be the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ with foci $S_1(3,0)$ and $S_2(-3,0)$. Let C be the parabola whose vertex is the centre of E and whose focus is S_1. If P is a point of intersection of E and C, then the value of $PS_1 - PS_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{25}+\frac{y^2}{16}=1$ है जिसकी नाभियाँ $S_1(3,0)$ तथा $S_2(-3,0)$ हैं। माना कि C वह परवलय है जिसका शीर्ष E का केन्द्र है तथा जिसकी नाभि S_1 है। यदि P, E तथा C का एक प्रतिच्छेदन बिन्दु है, तो $PS_1 - PS_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For E: $a=5$, $b=4$, $c=\sqrt{25-16}=3$, so the centre is $O(0,0)$ and the foci are $(\pm 3,0)$.

The parabola C has vertex O and focus $S_1(3,0)$, so its axis is the x-axis, $a=3$, $C:\ y^2=12x$, and its directrix is $x=-3$ (which passes through S_2).

Points common to E and C: $\frac{x^2}{25}+\frac{12x}{16}=1$, i.e. $4x^2+75x-100=0$, i.e. $(4x-5)(x+20)=0$. On C, $x\geq 0$, so $x=\frac{5}{4}$ and $y^2=15$: $P=\left(\frac{5}{4},\pm\sqrt{15}\right)$.

Focal property of the parabola: PS_1 equals the distance of P from the directrix $x=-3$, so $PS_1=\frac{5}{4}+3=\frac{17}{4}$. (Check with the ellipse: $PS_1=a-ex=5-\frac{3}{5}\cdot\frac{5}{4}=\frac{17}{4}$.)

Focal property of the ellipse: $PS_1+PS_2=2a=10$, so $PS_2=10-\frac{17}{4}=\frac{23}{4}$.

Hence $PS_1-PS_2=\frac{17}{4}-\frac{23}{4}=-\frac{3}{2}=-1.5$, the same for both points of intersection.

Final Answer: -1.50

Question 06

English

Let H be the hyperbola with centre at the origin and transverse axis along the x-axis whose eccentricity is 2 and whose two directrices are at a distance 2 from each other. Let S be the focus of H lying on the negative x-axis, and let C be the parabola whose focus is S and whose directrix is the directrix of H lying in the half-plane $x > 0$. Then the sum of the abscissae of all the points common to H and C is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि H वह अतिपरवलय है जिसका केन्द्र मूलबिन्दु पर है, अनुप्रस्थ अक्ष x-अक्ष के अनुदिश है, उत्केन्द्रता 2 है तथा जिसकी दोनों नियता रेखाओं के बीच की दूरी 2 है। माना कि S, H की वह नाभि है जो ऋणात्मक x-अक्ष पर स्थित है, तथा C वह परवलय है जिसकी नाभि S है और जिसकी नियता रेखा H की वह नियता रेखा है जो अर्धतल $x > 0$ में स्थित है। तब H तथा C के सभी उभयनिष्ठ बिन्दुओं के भुजों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $H: \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$. Its directrices are $x = \pm\dfrac{a}{e}$, so the distance between them is $\dfrac{2a}{e} = a = 2$ (as $e = 2$).

Then $c = ae = 4$ and $b^2 = a^2(e^2 - 1) = 4 \cdot 3 = 12$. So $H: \dfrac{x^2}{4} - \dfrac{y^2}{12} = 1$, the foci are $(\pm 4, 0)$, and the directrices are $x = \pm 1$.

$S = (-4, 0)$ and the directrix of C is $x = 1$. A point (x, y) lies on C when its distance from S equals its distance from $x = 1$: $(x+4)^2 + y^2 = (x-1)^2$, i.e. $y^2 = -10x - 15 = -10\left(x + \dfrac{3}{2}\right)$. (Vertex $\left(-\dfrac{3}{2}, 0\right)$, opening to the left; on C, $x \le -\dfrac{3}{2}$.)

Common points: from H, $3x^2 - y^2 = 12$. Substituting $y^2 = -10x - 15$: $3x^2 + 10x + 15 = 12$, i.e. $3x^2 + 10x + 3 = 0$, i.e. $(3x+1)(x+3) = 0$.

$x = -\dfrac{1}{3}$ gives $y^2 = \dfrac{10}{3} - 15 < 0$: no real point (indeed $-\dfrac{1}{3} > -\dfrac{3}{2}$ is outside C).

$x = -3$ gives $y^2 = 15$: the two points $(-3, \sqrt{15})$ and $(-3, -\sqrt{15})$. Check on H: $\dfrac{9}{4} - \dfrac{15}{12} = 1$.

Sum of the abscissae of all common points $= (-3) + (-3) = -6$.

Final Answer: -6

Question 07

English

Let C be the parabola whose focus is $S(1,2)$ and whose directrix is the line $x=-2$. Let V be the vertex of C and let L be the end of the latus rectum of C that lies in the first quadrant. The circle described on VL as a diameter meets the y-axis at the points A and B. If O is the origin, then the value of $OA \cdot OB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वह परवलय है जिसकी नाभि $S(1,2)$ है तथा जिसकी नियता रेखा $x=-2$ है। माना कि V, C का शीर्ष है तथा L, C के नाभिलम्ब का वह सिरा है जो प्रथम चतुर्थांश में स्थित है। VL को व्यास मानकर खींचा गया वृत्त y-अक्ष को बिन्दुओं A तथा B पर मिलता है। यदि O मूलबिन्दु है, तो $OA \cdot OB$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The axis of C is the line through $S(1,2)$ perpendicular to the directrix $x=-2$, i.e. $y=2$. It meets the directrix at $Z(-2,2)$.

The vertex is the midpoint of ZS: $V=\left(-\frac{1}{2},2\right)$, and $a=VS=\frac{3}{2}$. (So $C:\ (y-2)^2=6\left(x+\frac{1}{2}\right)$.)

The latus rectum is the focal chord perpendicular to the axis, of length $4a=6$, so its ends are $(1,2+3)=(1,5)$ and $(1,2-3)=(1,-1)$. The end in the first quadrant is $L(1,5)$. Check: $(5-2)^2=9=6\left(1+\frac{1}{2}\right)$.

The circle with $V\left(-\frac{1}{2},2\right)$ and $L(1,5)$ as ends of a diameter is $\left(x+\frac{1}{2}\right)(x-1)+(y-2)(y-5)=0$, i.e. $x^2+y^2-\frac{1}{2}x-7y+\frac{19}{2}=0$.

On the y-axis ($x=0$): $y^2-7y+\frac{19}{2}=0$. Its discriminant is $49-38=11>0$, so there are two points A and B, with $y_A+y_B=7$ and $y_Ay_B=\frac{19}{2}$. Both roots are positive, so $OA=y_A$ and $OB=y_B$.

Hence $OA \cdot OB=\frac{19}{2}=9.5$. (This is also the power of O with respect to the circle, the constant term $\frac{19}{2}$.)

Final Answer: 9.50

Question 08

English

Let E be the ellipse $x^2 + 9y^2 - 2x - 36y + 28 = 0$. Let C be the parabola, with axis parallel to the y-axis, that passes through both ends of the major axis of E and through the upper end of the minor axis of E. If the focus of C is the point (α, β), then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $x^2 + 9y^2 - 2x - 36y + 28 = 0$ है। माना कि C वह परवलय है, जिसका अक्ष y-अक्ष के समान्तर है, जो E के दीर्घ अक्ष के दोनों सिरों से तथा E के लघु अक्ष के ऊपरी सिरे से होकर जाता है। यदि C की नाभि बिन्दु (α, β) है, तो β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Completing squares: $x^2-2x+1+9(y^2-4y+4) = -28+1+36 = 9$, so $E: \ \dfrac{(x-1)^2}{9} + \dfrac{(y-2)^2}{1} = 1$.

Centre $(1, 2)$, semi-major axis 3 along the horizontal line $y = 2$, semi-minor axis 1 along $x = 1$.

Ends of the major axis: $(1 \pm 3, 2)$, i.e. $(-2, 2)$ and $(4, 2)$. Ends of the minor axis: $(1, 2 \pm 1)$; the upper end is $(1, 3)$.

A parabola with axis parallel to the y-axis is $y = px^2 + qx + r$. It takes the same value 2 at $x = -2$ and $x = 4$, so its axis is $x = \dfrac{-2+4}{2} = 1$. The point $(1, 3)$ lies on the axis, so it is the vertex: $C: \ (x-1)^2 = -4A(y-3)$.

Through $(4, 2)$: $9 = -4A(2-3) = 4A$, so $A = \dfrac{9}{4} > 0$ and $C: \ (x-1)^2 = -9(y-3)$ opens downward.

The focus is at distance A below the vertex: $\left(1, 3 - \dfrac{9}{4}\right) = \left(1, \dfrac{3}{4}\right)$. So $\alpha = 1$, $\beta = \dfrac{3}{4} = 0.75$.

Final Answer: 0.75

Question 09

English

Each chord of the parabola $y^2 = 12x$ that passes through the point $A(15, 6)$ bounds, together with the parabola, a region. Then the least possible area of such a region is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

परवलय $y^2 = 12x$ की प्रत्येक जीवा, जो बिन्दु $A(15, 6)$ से होकर जाती है, परवलय के साथ एक क्षेत्र परिबद्ध करती है। तब ऐसे क्षेत्र का न्यूनतम सम्भव क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $6^2 = 36 < 12 \cdot 15 = 180$, the point $A(15, 6)$ is inside the parabola, so every line through A other than $y = 6$ (parallel to the axis, meeting the curve once) cuts the parabola in a chord.

Such a line is $x - 15 = m(y - 6)$, i.e. $x = my + 15 - 6m$, $m \in \mathbb{R}$.

Substituting $x = \dfrac{y^2}{12}$: $y^2 - 12my - 12(15 - 6m) = 0$. If $y_1 < y_2$ are its roots, $(y_2 - y_1)^2 = (12m)^2 + 48(15 - 6m) = 144m^2 - 288m + 720 = 144\left[(m - 1)^2 + 4\right]$.

The region between the chord and the parabola has area $\displaystyle\int_{y_1}^{y_2} \left(my + 15 - 6m - \frac{y^2}{12}\right) dy = \frac{1}{12}\int_{y_1}^{y_2} (y - y_1)(y_2 - y)\, dy = \frac{(y_2 - y_1)^3}{72}$.

This increases with $y_2 - y_1$, which is least when $m = 1$: $(y_2 - y_1)^2 = 576$, $y_2 - y_1 = 24$. (At $m = 1$ the chord is bisected by A: $\dfrac{y_1 + y_2}{2} = 6m = 6$.)

For $m = 1$ the chord is $y = x - 9$, meeting the parabola at $y = -6$ and $y = 18$, i.e. at $(3, -6)$ and $(27, 18)$; its length is $\sqrt{24^2 + 24^2} = 24\sqrt{2}$.

By Archimedes' result the segment is $\dfrac{2}{3}$ of the parallelogram formed by the chord and the tangent parallel to it. The tangent of slope 1 to $y^2 = 12x$ touches at $\left(\dfrac{3}{1}, \dfrac{6}{1}\right) = (3, 6)$; its distance from $x - y - 9 = 0$ is $\dfrac{|3 - 6 - 9|}{\sqrt{2}} = 6\sqrt{2}$.

Least area $= \dfrac{2}{3} \cdot 24\sqrt{2} \cdot 6\sqrt{2} = \dfrac{2}{3} \cdot 288 = 192$ (the same as $\dfrac{24^3}{72}$).

Final Answer: 192

Question 10

English

Let S be the set of all real numbers λ for which the point $(\lambda, 2\lambda+1)$ lies inside the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and outside the parabola $y^2=4(x+1)$. Then the sum of all the integers in S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं λ का समुच्चय है जिनके लिए बिन्दु $(\lambda, 2\lambda+1)$ दीर्घवृत्त $\frac{x^2}{25}+\frac{y^2}{16}=1$ के अन्दर तथा परवलय $y^2=4(x+1)$ के बाहर स्थित है। तब S में स्थित सभी पूर्णांकों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A point (x_1, y_1) is inside the ellipse when $\frac{x_1^2}{25}+\frac{y_1^2}{16}-1<0$, and outside the parabola $y^2=4(x+1)$ when $y_1^2-4(x_1+1)>0$.

Ellipse: $\frac{\lambda^2}{25}+\frac{(2\lambda+1)^2}{16}<1 \iff 16\lambda^2+25(4\lambda^2+4\lambda+1)<400 \iff 116\lambda^2+100\lambda-375<0.$

The roots are $\lambda=\frac{-100\pm\sqrt{10000+174000}}{232}=\frac{-25\pm10\sqrt{115}}{58}$, i.e. about -2.280 and 1.418. So the condition is $-2.280<\lambda<1.418$ (approximately).

Parabola: $(2\lambda+1)^2-4(\lambda+1)=4\lambda^2-3>0 \iff \lambda<-\frac{\sqrt{3}}{2}$ or $\lambda>\frac{\sqrt{3}}{2}$ ($\frac{\sqrt{3}}{2}\approx0.866$).

Hence $S=\left(\frac{-25-10\sqrt{115}}{58},-\frac{\sqrt{3}}{2}\right)\cup\left(\frac{\sqrt{3}}{2},\frac{-25+10\sqrt{115}}{58}\right)$.

Integers in S: -2 (point $(-2,-3)$: $\frac{4}{25}+\frac{9}{16}<1$ and $9-(-4)>0$), -1 (point $(-1,-1)$: inside E, $1-0>0$), and 1 (point $(1,3)$: $\frac{1}{25}+\frac{9}{16}<1$, $9-8>0$). $\lambda=0$ fails ($(0,1)$: $1-4<0$, inside the parabola), $\lambda=2$ fails ($(2,5)$ is outside E) and $\lambda=-3$ fails ($(-3,-5)$ is outside E).
Sum $=(-2)+(-1)+1=-2$.

Final Answer: -2

Question 11

English

A line L touches the parabola $y^2 = 24x$ and also touches the curve $C: \ y = x^3$ at the point Q. The line L meets C again at the point R. Then the value of $(QR)^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक रेखा L परवलय $y^2 = 24x$ को स्पर्श करती है तथा वक्र $C: \ y = x^3$ को भी बिन्दु Q पर स्पर्श करती है। रेखा L, वक्र C को पुनः बिन्दु R पर मिलती है। तब $(QR)^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let L touch C at $Q(s, s^3)$. Since $\dfrac{dy}{dx} = 3x^2$, the tangent to C at Q is $y - s^3 = 3s^2(x - s)$, i.e. $y = 3s^2x - 2s^3$.

If $s = 0$, this is $y = 0$, the axis of the parabola, which meets $y^2 = 24x$ only once without touching it. So $s \neq 0$ and the slope $m = 3s^2 > 0$.

For $y^2 = 24x$ ($4a = 24$, $a = 6$), the line $y = mx + c$ with $m \neq 0$ is a tangent if and only if $c = \dfrac{a}{m} = \dfrac{6}{m}$.

So $-2s^3 = \dfrac{6}{3s^2} = \dfrac{2}{s^2}$, i.e. $s^5 = -1$, whose only real root is $s = -1$.

Hence $Q = (-1, -1)$ and $L: \ y = 3x + 2$. Check with the parabola: $(3x + 2)^2 = 24x \iff (3x - 2)^2 = 0$, a double root, so L touches it at $\left(\dfrac{2}{3}, 4\right)$.

Points common to L and C: $x^3 = 3x + 2 \iff x^3 - 3x - 2 = 0 \iff (x + 1)^2(x - 2) = 0$. The double root $x = -1$ is Q; the other is $x = 2$, so $R = (2, 8)$.

$(QR)^2 = (2 + 1)^2 + (8 + 1)^2 = 9 + 81 = 90$.

Final Answer: 90

Question 12

English

Let PQ be a focal chord of the parabola $y^2 = 36x$. The region bounded by the parabola and the tangents to it at P and Q has area 500. If the slope of PQ is negative, then the slope of PQ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि PQ परवलय $y^2 = 36x$ की एक नाभि जीवा है। परवलय तथा P और Q पर इसकी स्पर्श रेखाओं से परिबद्ध क्षेत्र का क्षेत्रफल 500 है। यदि PQ की प्रवणता ऋणात्मक है, तो PQ की प्रवणता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $y^2 = 36x$, $a = 9$. Let $P = (9t_1^2, 18t_1)$, $Q = (9t_2^2, 18t_2)$. The chord PQ passes through the focus $(9, 0)$ exactly when $t_1 t_2 = -1$.

The tangents $t_1 y = x + 9t_1^2$ and $t_2 y = x + 9t_2^2$ meet at $T = (9t_1 t_2,\, 9(t_1 + t_2)) = (-9,\, 9(t_1 + t_2))$.

Area of triangle PTQ: $\dfrac{1}{2}\left|\begin{vmatrix} 9t_1^2 & 18t_1 & 1 \\ 9t_2^2 & 18t_2 & 1 \\ 9t_1 t_2 & 9(t_1+t_2) & 1 \end{vmatrix}\right| = \dfrac{81}{2}|t_1 - t_2|^3$.

The region between the chord PQ and the arc is a parabolic segment; by Archimedes it is $\dfrac{2}{3}$ of the triangle formed by the chord and the tangents at its ends, i.e. $\dfrac{81}{3}|t_1 - t_2|^3$.

The region bounded by the parabola and the two tangents is the triangle minus the segment: $\dfrac{81}{2}D^3 - \dfrac{81}{3}D^3 = \dfrac{81}{6}D^3$, where $D = |t_1 - t_2|$.

$\dfrac{81}{6}D^3 = 500 \Rightarrow D^3 = \dfrac{3000}{81} = \dfrac{1000}{27} \Rightarrow D = \dfrac{10}{3}$.

With $t_2 = -\dfrac{1}{t_1}$: $D = \left|t_1 + \dfrac{1}{t_1}\right| = |t_1| + \dfrac{1}{|t_1|} = \dfrac{10}{3}$, so $|t_1| = 3$ or $|t_1| = \dfrac{1}{3}$.

Slope of $PQ = \dfrac{18t_1 - 18t_2}{9t_1^2 - 9t_2^2} = \dfrac{2}{t_1 + t_2} = \dfrac{2}{t_1 - \frac{1}{t_1}} = \dfrac{2t_1}{t_1^2 - 1}$. For $t_1 = 3$: $\dfrac{6}{8} = \dfrac{3}{4}$; for $t_1 = -3$: $-\dfrac{3}{4}$; for $t_1 = \pm\dfrac{1}{3}$: $\mp\dfrac{3}{4}$ (the same chords with P and Q interchanged).

So the slope is $\pm\dfrac{3}{4}$, and the negative value is $-\dfrac{3}{4} = -0.75$.

Final Answer: -0.75

Question 13

English

Let C be the parabola $y^2 = 12x$ and let E be the ellipse $\frac{x^2}{\alpha^2} + \frac{y^2}{16} = 1$, where $\alpha > 4$. Suppose that there is a point Q on E such that the tangent to E at Q and the normal to E at Q both touch C. Then the eccentricity of E is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C परवलय $y^2 = 12x$ है तथा E दीर्घवृत्त $\frac{x^2}{\alpha^2} + \frac{y^2}{16} = 1$ है, जहाँ $\alpha > 4$ है। माना कि E पर एक बिन्दु Q ऐसा है कि Q पर E की स्पर्श रेखा तथा Q पर E का अभिलम्ब, दोनों C को स्पर्श करते हैं। तब E की उत्केन्द्रता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The tangent and the normal to E at Q are two perpendicular lines through Q. Both touch C, so Q is a point from which two perpendicular tangents can be drawn to C. The locus of such points (the director circle of a parabola) is its directrix, here $x = -3$.

So $Q = (-3, k)$. If $k = 0$, $Q = (-3, 0)$ would need $\alpha = 3$, contrary to $\alpha > 4$; hence $k \neq 0$, and the tangent to E at Q is not vertical.

Tangent to E at Q: $\frac{-3x}{\alpha^2} + \frac{ky}{16} = 1$, i.e. $y = \frac{48}{\alpha^2 k}x + \frac{16}{k}$: slope $m = \frac{48}{\alpha^2 k}$, intercept $c = \frac{16}{k}$.

For $y^2 = 12x$ $(a = 3)$ the line $y = mx + c$ is a tangent iff $c = \frac{3}{m}$: $\frac{16}{k} = \frac{3\alpha^2 k}{48} = \frac{\alpha^2 k}{16}$, so $k^2 = \frac{256}{\alpha^2}$.

Q lies on E: $\frac{9}{\alpha^2} + \frac{k^2}{16} = 1$, i.e. $\frac{9}{\alpha^2} + \frac{16}{\alpha^2} = 1$, so $\alpha^2 = 25$ and $k = \pm\frac{16}{5}$.

Check the normal: at $Q\left(-3, \frac{16}{5}\right)$ the tangent has slope $\frac{48}{25 \cdot \frac{16}{5}} = \frac{3}{5}$, so the normal is $y - \frac{16}{5} = -\frac{5}{3}(x + 3)$, i.e. $y = -\frac{5}{3}x - \frac{9}{5}$; and $\frac{3}{m} = \frac{3}{-5/3} = -\frac{9}{5}$, so it touches C too.

Eccentricity of E: $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5} = 0.6$. (The foci of E are $(\pm 3, 0)$, so $x = -3$ is the latus-rectum line of E and Q is an end of the latus rectum.)

Final Answer: 0.60

Question 14

English

Let C be the curve $y = x^3 + 3x^2 + 5x + 8$. The normal to C at its point of inflection is a normal to the parabola $y^2 = 16x$ at a point P, and it meets the parabola again at the point Q. If O is the origin, then the area of the triangle OPQ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वक्र $y = x^3 + 3x^2 + 5x + 8$ है। C के नति परिवर्तन बिन्दु पर C का अभिलम्ब, परवलय $y^2 = 16x$ के किसी बिन्दु P पर अभिलम्ब है, तथा यह परवलय को पुनः बिन्दु Q पर मिलता है। यदि O मूलबिन्दु है, तो त्रिभुज OPQ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y' = 3x^2 + 6x + 5$ and $y'' = 6x + 6$. y'' changes sign at $x = -1$, so the point of inflection is $(-1,\ -1 + 3 - 5 + 8) = (-1, 5)$, where $y' = 3 - 6 + 5 = 2$.

The normal to C there has slope $-\dfrac{1}{2}$: $y - 5 = -\dfrac{1}{2}(x + 1)$, i.e. $y = -\dfrac{1}{2}x + \dfrac{9}{2}$, or $x + 2y = 9$.

For $y^2 = 16x$, $a = 4$; the normal at $(4t^2, 8t)$ is $y = -tx + 8t + 4t^3$, with slope $-t$. A normal of slope $-\dfrac{1}{2}$ has $t = \dfrac{1}{2}$: $y = -\dfrac{1}{2}x + 4 + \dfrac{1}{2} = -\dfrac{1}{2}x + \dfrac{9}{2}$, which is exactly the line above. So $P = (4t^2, 8t) = (1, 4)$.

The normal at t meets the parabola again at the parameter t' with $t + t' = -\dfrac{2}{t}$ (from the chord PQ having slope $\dfrac{2}{t + t'} = -t$): $t' = -\dfrac{1}{2} - 4 = -\dfrac{9}{2}$, so $Q = \left(4 \cdot \dfrac{81}{4},\ 8 \cdot \left(-\dfrac{9}{2}\right)\right) = (81, -36)$.

Check: $x + 2y = 81 - 72 = 9$, and $(-36)^2 = 1296 = 16 \cdot 81$.

Area of triangle $OPQ = \dfrac{1}{2}|x_P y_Q - x_Q y_P| = \dfrac{1}{2}|1 \cdot (-36) - 81 \cdot 4| = \dfrac{1}{2} \cdot 360 = 180$.

Final Answer: 180

Question 15

English

Three normals are drawn from a point $N(h, k)$ to the parabola $y^2 = 6x$, and their feet are the points P, Q and R. The tangents to the parabola at P and Q meet at the point $T\left(-6, \frac{3}{2}\right)$. Then the value of h is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक बिन्दु $N(h, k)$ से परवलय $y^2 = 6x$ पर तीन अभिलम्ब खींचे जाते हैं, जिनके पाद बिन्दु P, Q तथा R हैं। P तथा Q पर परवलय की स्पर्श रेखाएँ बिन्दु $T\left(-6, \frac{3}{2}\right)$ पर मिलती हैं। तब h का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $y^2 = 6x$, $4a = 6$, $a = \frac{3}{2}$. Let the feet be P, Q, R with parameters t_1, t_2, t_3, i.e. points $\left(\frac{3}{2}t^2, 3t\right)$.

The tangents at t_1 and t_2 are $t_1 y = x + at_1^2$ and $t_2 y = x + at_2^2$; they meet at $(at_1t_2,\ a(t_1 + t_2))$. Equating with $T\left(-6, \frac{3}{2}\right)$: $\frac{3}{2}t_1t_2 = -6$ and $\frac{3}{2}(t_1 + t_2) = \frac{3}{2}$, so $t_1t_2 = -4$, $t_1 + t_2 = 1$. (t_1, t_2 are the roots of $t^2 - t - 4 = 0$, real and distinct.)

The normal at t is $y = -tx + 2at + at^3$. It passes through $N(h, k)$ when $at^3 + (2a - h)t - k = 0$. This cubic has the roots t_1, t_2, t_3 (the three co-normal feet), and it has no t^2 term.

So $t_1 + t_2 + t_3 = 0$, giving $t_3 = -1$, and $t_1t_2 + t_2t_3 + t_3t_1 = \frac{2a - h}{a}$.

$t_1t_2 + t_3(t_1 + t_2) = -4 + (-1)(1) = -5$, so $2a - h = -5a$ and $h = 7a = \frac{21}{2}$.

Also $t_1t_2t_3 = \frac{k}{a}$ gives $k = \frac{3}{2} \cdot 4 = 6$. Check with the normal at $t_3 = -1$: $y = x - 3 - \frac{3}{2} = x - \frac{9}{2}$, and at $x = \frac{21}{2}$, $y = 6$.

Hence $h = 10.5$.

Final Answer: 10.50

Question 16

English

The set of all real numbers h for which three distinct normals can be drawn to the parabola $y^2 = 4x$ from the point $(h, h-2)$ is the interval (α, ∞). Then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

उन सभी वास्तविक संख्याओं h का समुच्चय, जिनके लिए बिन्दु $(h, h-2)$ से परवलय $y^2 = 4x$ पर तीन भिन्न अभिलम्ब खींचे जा सकते हैं, अन्तराल (α, ∞) है। तब α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The normal to $y^2 = 4x$ at $(t^2, 2t)$ is $y = -tx + 2t + t^3$. It passes through $(h, h-2)$ when $t^3 + (2-h)t - (h-2) = 0$. Put $u = h - 2$: $g(t) = t^3 - ut - u = 0$.

Distinct feet give distinct normals (different slopes $-t$), so three distinct normals exist exactly when g has three distinct real roots.

If $u \le 0$: $g'(t) = 3t^2 - u \ge 0$, with equality at most at one point, so g is strictly increasing and has exactly one real root.

If $u > 0$: $g'(t) = 0$ at $t = \pm c$, $c = \sqrt{\dfrac{u}{3}}$; g has a local maximum at $-c$ and a local minimum at c.

Three distinct real roots $\iff g(-c) > 0 > g(c) \iff g(c)g(-c) < 0$ (since $g(-c) > g(c)$).

Using $c^2 = \dfrac{u}{3}$: $g(c) = \dfrac{u}{3}c - uc - u = -\dfrac{2u}{3}c - u$ and $g(-c) = \dfrac{2u}{3}c - u$.

$g(c)g(-c) = u^2 - \dfrac{4u^2}{9}c^2 = u^2 - \dfrac{4u^3}{27} = u^2\left(1 - \dfrac{4u}{27}\right)$. This is negative exactly when $u > \dfrac{27}{4}$.

At $u = \dfrac{27}{4}$ the cubic has a double root (only two distinct normals). So the condition is $h - 2 > \dfrac{27}{4}$, i.e. $h > \dfrac{35}{4}$, and $\alpha = \dfrac{35}{4} = 8.75$.

Final Answer: 8.75

Question 17

English

Let C be the parabola $y^2 - 2y - 5x + 6 = 0$, and let $P(\alpha, \beta)$ be a point from which the two tangents drawn to C are perpendicular to each other and whose chord of contact with respect to C is parallel to the line $2x + y = 0$. Then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C परवलय $y^2 - 2y - 5x + 6 = 0$ है, तथा $P(\alpha, \beta)$ एक ऐसा बिन्दु है जिससे C पर खींची गई दोनों स्पर्श रेखाएँ परस्पर लम्बवत् हैं और C के सापेक्ष जिसकी स्पर्श जीवा रेखा $2x + y = 0$ के समान्तर है। तब $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Completing the square: $y^2 - 2y + 1 = 5x - 5$, so $C: \ (y-1)^2 = 5(x-1)$. With $X = x - 1$, $Y = y - 1$ this is $Y^2 = 4AX$, $A = \dfrac{5}{4}$; vertex $(1, 1)$, axis $y = 1$, directrix $X = -A$, i.e. $x = 1 - \dfrac{5}{4} = -\dfrac{1}{4}$.

The locus of points from which two perpendicular tangents can be drawn to a parabola (its director circle) is the directrix. Hence $\alpha = -\dfrac{1}{4}$.

The chord of contact of (X_1, Y_1) for $Y^2 = 4AX$ is $YY_1 = 2A(X + X_1)$. Here $X_1 = \alpha - 1$, $Y_1 = \beta - 1$: $(\beta - 1)(y - 1) = \dfrac{5}{2}(x - 1 + \alpha - 1)$.

Its slope is $\dfrac{5}{2(\beta - 1)}$ (note $\beta \neq 1$, otherwise the chord would be vertical). Parallel to $2x + y = 0$ means slope -2: $\dfrac{5}{2(\beta - 1)} = -2$, so $\beta - 1 = -\dfrac{5}{4}$, $\beta = -\dfrac{1}{4}$.

Check: $P\left(-\dfrac{1}{4}, -\dfrac{1}{4}\right)$ is outside C, since $\left(-\dfrac{5}{4}\right)^2 - 5\left(-\dfrac{5}{4}\right) > 0$; the tangents from P have slopes m with $m^2 - m - 1 = 0$, product -1.

$\alpha + \beta = -\dfrac{1}{4} - \dfrac{1}{4} = -\dfrac{1}{2} = -0.5$.

Final Answer: -0.50

Question 18

English

Let E be the ellipse $\frac{x^2}{32} + \frac{y^2}{8} = 1$ and let C be the parabola $y^2 = 4x$. Let P be a point such that the polar of P with respect to E touches C and the polar of P with respect to C touches E. If O is the origin, then the value of OP^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{32} + \frac{y^2}{8} = 1$ है तथा C परवलय $y^2 = 4x$ है। माना कि P एक ऐसा बिन्दु है कि E के सापेक्ष P की ध्रुवी, C को स्पर्श करती है तथा C के सापेक्ष P की ध्रुवी, E को स्पर्श करती है। यदि O मूलबिन्दु है, तो OP^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $P = (h, k)$. The polar of P with respect to E is $\frac{hx}{32} + \frac{ky}{8} = 1$ (for $P = O$ there is no polar line, so $P \neq O$), and with respect to C it is $ky = 2(x + h)$.

If $k = 0$, then $h \neq 0$ and the polar with respect to E is $x = \frac{32}{h}$, a vertical line; the only vertical tangent of C is $x = 0$, so this is impossible. Hence $k \neq 0$.

Polar w.r.t. E: $y = -\frac{h}{4k}x + \frac{8}{k}$. A line $y = mx + c$ touches $y^2 = 4x$ iff $m \neq 0$ and $c = \frac{1}{m}$, i.e. $cm = 1$: $\frac{8}{k} \cdot \left(-\frac{h}{4k}\right) = 1$, so $k^2 = -2h$ (and $h < 0$).

Polar w.r.t. C: $y = \frac{2}{k}x + \frac{2h}{k}$. A line $y = mx + c$ touches $\frac{x^2}{32} + \frac{y^2}{8} = 1$ iff $c^2 = 32m^2 + 8$: $\frac{4h^2}{k^2} = \frac{128}{k^2} + 8$, i.e. $h^2 = 32 + 2k^2$.

Substituting $k^2 = -2h$: $h^2 + 4h - 32 = 0$, i.e. $(h + 8)(h - 4) = 0$. $h = 4$ would give $k^2 = -8$; so $h = -8$ and $k^2 = 16$, $P = (-8, \pm 4)$.

Check for $P(-8, 4)$: polar w.r.t. E is $y = \frac{x}{2} + 2$, and $c = 2 = \frac{1}{1/2}$; polar w.r.t. C is $y = \frac{x}{2} - 4$, and $c^2 = 16 = 32 \cdot \frac{1}{4} + 8$.

$OP^2 = (-8)^2 + 4^2 = 80$ for both points.

Final Answer: 80

Question 19

English

A ray of light travelling along the line $y = 16$ in the direction of decreasing x strikes the parabolic mirror $y^2 = 16x$ at the point P. After reflection at P, the ray strikes the mirror again at the point Q. The tangents to the mirror at P and Q meet at the point T, and the normals to the mirror at P and Q meet at the point N. Then the area of the quadrilateral $PTQN$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

रेखा $y = 16$ के अनुदिश x के घटने की दिशा में चलती हुई एक प्रकाश किरण परवलयिक दर्पण $y^2 = 16x$ से बिन्दु P पर टकराती है। P पर परावर्तन के बाद किरण दर्पण से पुनः बिन्दु Q पर टकराती है। P तथा Q पर दर्पण की स्पर्श रेखाएँ बिन्दु T पर मिलती हैं, तथा P और Q पर दर्पण के अभिलम्ब बिन्दु N पर मिलते हैं। तब चतुर्भुज $PTQN$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $y^2 = 16x$, $a = 4$, the focus is $S(4, 0)$ and the directrix is $x = -4$. The ray $y = 16$ meets the mirror where $16^2 = 16x$, i.e. at $P(16, 16)$, the point with parameter $t_1 = 2$ (points $(4t^2, 8t)$). It arrives from the side $x > 16$, i.e. inside the parabola.

Reflection property: the tangent at P makes equal angles with the line through P parallel to the axis and with PS; so a ray travelling parallel to the axis is reflected along PS. The reflected ray passes through the focus, so PQ is a focal chord and $t_1 t_2 = -1$: $t_2 = -\dfrac{1}{2}$, $Q = (1, -4)$.

Tangent at P: $t_1 y = x + at_1^2$, i.e. $2y = x + 16$, slope $\dfrac{1}{2}$. Tangent at Q: $-\dfrac{1}{2}y = x + 1$, slope -2. The product of the slopes is -1; they meet at $T = (at_1 t_2, a(t_1 + t_2)) = (-4, 6)$, on the directrix.

The normal at P is perpendicular to the tangent at P, and the normal at Q is perpendicular to the tangent at Q. Since the two tangents are perpendicular, the normal at P is parallel to the tangent at Q and the normal at Q is parallel to the tangent at P. So $PTQN$ is a rectangle with vertices in this order. (Directly: normal at P: $y - 16 = -2(x - 16)$; normal at Q: $y + 4 = \dfrac{1}{2}(x - 1)$; they meet at $N(21, 6)$.)

$PT = \sqrt{(16 + 4)^2 + (16 - 6)^2} = \sqrt{500} = 10\sqrt{5}$ and $QT = \sqrt{(1 + 4)^2 + (-4 - 6)^2} = \sqrt{125} = 5\sqrt{5}$.

Area of the rectangle $PTQN = PT \cdot QT = 10\sqrt{5} \cdot 5\sqrt{5} = 250$.

Final Answer: 250

Question 20

English

The parabola $y^2 = 2(x - h)$ touches the ellipse $\frac{x^2}{12} + \frac{y^2}{6} = 1$ at two distinct points. Then the value of h is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

परवलय $y^2 = 2(x - h)$ दीर्घवृत्त $\frac{x^2}{12} + \frac{y^2}{6} = 1$ को दो भिन्न बिन्दुओं पर स्पर्श करता है। तब h का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both curves are symmetric in the x-axis, so two distinct contact points are (x_1, y_1) and $(x_1, -y_1)$ with $y_1 \neq 0$. At a contact point the curves have the same tangent, hence the same normal, and this normal is not the x-axis.

Parabola $y^2 = 2(x - h)$: here $4A = 2$, $A = \frac{1}{2}$. Its subnormal is constant and equal to $2A = 1$: differentiating, $2y\frac{dy}{dx} = 2$, so the subnormal $y\frac{dy}{dx} = 1$, and the normal at (x_1, y_1) meets the axis at $(x_1 + 1, 0)$.

Ellipse $\frac{x^2}{12} + \frac{y^2}{6} = 1$: $e^2 = 1 - \frac{b^2}{a^2} = \frac{1}{2}$. The normal at (x_1, y_1), $\frac{12x}{x_1} - \frac{6y}{y_1} = 12 - 6$, meets the major axis at $x = \frac{6x_1}{12} = e^2 x_1 = \frac{x_1}{2}$.

The common normal meets the axis at one point: $x_1 + 1 = \frac{x_1}{2}$, so $x_1 = -2$.

From the ellipse, $y_1^2 = 6\left(1 - \frac{4}{12}\right) = 4$, so the contact points are $(-2, \pm 2)$. From the parabola, $4 = 2(-2 - h)$, so $h = -4$.

Check: with $h = -4$, eliminating y: $\frac{x^2}{12} + \frac{2(x+4)}{6} = 1 \iff x^2 + 4x + 4 = 0 \iff (x+2)^2 = 0$, a double root, so the curves touch at $(-2, \pm 2)$ and meet nowhere else.

Final Answer: -4